

Robust Perception in Adverse Weather: A Review of Multi-Sensor Fusion Strategies for Autonomous Driving

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Abstract: With the evolution of autonomous driving technology towards L4/L5 levels, the reliability of perception systems under adverse weather conditions—such as rain, snow, and dense fog—has emerged as a critical bottleneck restricting their large-scale commercial deployment. In complex "long-tail" scenarios, single-modality sensors are constrained by physical characteristics like scattering and attenuation, making it difficult to maintain stable perception capabilities. This paper presents a systematically review robust multi-sensor fusion strategies tailored for adverse weather environments. First, we dissect the asymmetric failure mechanisms of different sensors in extreme environmental media from a physical perspective and summarize the latest specialized benchmarks for adverse weather. Subsequently, we delve into the evolutionary trends of fusion architectures, with a specific focus on deep fusion based on unified Bird's-Eye-View (BEV) feature spaces, Transformer-based soft-attention association mechanisms, and environment-aware adaptive weight allocation strategies. Furthermore, this paper analyzes the engineering barriers constraining real-world deployment, specifically addressing the trade-offs between computational efficiency and real-time performance, the necessity of SOTIF certification, and weather-induced system integration failures. Finally, we discuss the application prospects of 4D imaging radar and Vehicle-to-Everything (V2X) cooperative perception in resolving all-weather perception limits.

Keywords: *Autonomous Driving, Multi-Sensor Fusion, Adverse Weather, Robust Perception, Adaptive Weighting, Engineering Challenges, SOTIF*

I. INTRODUCTION

As autonomous driving technology evolves from Advanced Driver Assistance Systems (L2) to Full Self-Driving (L5), the reliability of perception systems has emerged as a critical bottleneck constraining commercial deployment. While state-of-the-art perception algorithms demonstrate exceptional performance in structured environments with favorable conditions, they remain vulnerable to complex "long-tail" scenarios^[1]. Empirical evidence indicates that adverse weather is a primary factor contributing to increased disengagement rates and safety incidents in autonomous vehicles. Consequently, enhancing the robustness of perception systems under adverse weather conditions has become an urgent priority for both the academic and industrial communities^[2].

In the context of this study, "adverse weather" refers not merely to overcast skies or light precipitation, but to meteorological conditions that significantly interfere with the

physical imaging characteristics of sensors. These primarily encompass three categories: precipitation (rain, snow) which causes optical refraction and occlusion; aerosols (dense fog, sandstorms) which reduce visibility and induce scattering; and extreme lighting conditions (low illumination or high glare)^[3]. These environmental factors not only attenuate the signal-to-noise ratio (SNR) of sensors but also introduce stochastic noise—for instance, raindrops can cause severe signal distortion on camera lenses^[4]—thereby rendering traditional algorithms ineffective.

Single-modality sensors exhibit inherent physical limitations under adverse weather conditions and cannot independently satisfy the safety redundancy requirements of L4/L5 autonomous driving. Specifically, cameras, while cost-effective and rich in semantic information, suffer significant performance degradation under low light, glare, or physical occlusion by rain and snow. LiDAR, as an active sensor, excels in ranging accuracy and 3D modeling but is susceptible to absorption and Mie scattering caused by rain and fog particles, leading to drastic reductions in detection range and a surge in point cloud noise^[5]. Conversely, while millimeter-wave radar offers all-weather penetrability and robustness against rain and fog, its inherent sparsity and limited angular resolution constrain its object classification capabilities^[5].

In summary, no single sensor is capable of handling all weather conditions and scenarios. Given that different sensors exhibit "asymmetric failure" characteristics under adverse weather^[6], multi-sensor information fusion has become the inevitable technical route. By integrating the complementary strengths of different modalities, fusion technology effectively increases system redundancy and mitigates the risks associated with single-source failure^[2]. Constructing a perception architecture based on multi-sensor fusion is, therefore, the core technological path to solving the robustness challenges of autonomous driving in extreme environments^[5].

II. SENSOR FAILURE MECHANISMS AND DATA BENCHMARKS UNDER ADVERSE WEATHER

A. Analysis of Physical Failure Mechanisms

The robustness of autonomous driving perception systems depends largely on the performance of sensors across different environmental media. Electromagnetic waves of different wavelengths exhibit distinct physical attenuation characteristics

when penetrating atmospheric precipitation particles (rain, snow) and suspended aerosols (fog, dust) [5].

Visual Sensors (Camera): Operating in the visible light spectrum (approx. 400–700 nm), cameras are highly susceptible to interference from illumination changes and environmental media. Water droplets adhering to the lens surface cause non-linear optical distortion and local occlusion, a phenomenon known as the "water-drop effect" [7]. In dense fog environments, suspended particulates cause light scattering, significantly degrading image contrast and visibility, which leads to the failure of texture-based detection algorithms [3].

LiDAR: Typically utilizing wavelengths of 905 nm or 1550 nm, LiDAR is vulnerable to weather particles. Since the size of fog and rain droplets (1–100 μm) is often close to or larger than the laser wavelength, the laser beam undergoes severe scattering and absorption during propagation according to Mie Scattering theory [5]. This not only results in range reduction but also induces backscatter from suspended particles, generating a large amount of random "ghost noise" in the point cloud and causing the system to misinterpret fog as obstacles [6].

Millimeter-wave Radar (Radar): Typically operating in the 77 GHz band (wavelength approx. 3.9 mm), radar follows Rayleigh Scattering laws because its wavelength is much larger than the size of fog droplets. Consequently, it possesses strong penetration capabilities through rain and fog. However, under heavy rain or wet road conditions, radar is susceptible to multipath effects and rain clutter, which degrades the Signal-to-Noise Ratio (SNR) [8].

B. Asymmetric Sensor Failure

Bijelic et al. [6] proposed the concept of "Asymmetric Sensor Failure," which serves as the theoretical cornerstone for multi-sensor fusion. This concept posits that under the same adverse weather condition, sensors of different modalities do not fail simultaneously or in the same manner. For instance, in dense fog, LiDAR and cameras may be completely "blinded," whereas radar can still effectively detect target distances. Conversely, in unlit nighttime environments, cameras lose effectiveness while LiDAR and radar remain unaffected. This complementary failure pattern allows the system to maintain stable and reliable perception via fusion algorithms by relying on functional modalities when a single source fails.

C. Benchmarks for Adverse Weather Perception

To enhance the robustness of autonomous perception systems, training stable models requires high-quality annotated data. Traditional datasets (e.g., KITTI, nuScenes) focus primarily on favorable weather scenarios and lack coverage of extreme environmental conditions. In recent years, the academic community has released a series of datasets dedicated to adverse weather.

RADIATE [8] is the first dataset containing high-resolution mechanical scanning radar data in adverse weather, covering seven scenarios including rain, fog, and snow, verifying the critical role of radar under conditions where vision fails. CADC [9] focuses on winter snowstorm driving environments, providing data on snow-covered roads and lane markings occluded by snow. SeeingThroughFog (DENSE) [6] provides multi-modal

data in dense fog environments and introduces gated camera data, providing support for research on penetrating perception in fog.

Furthermore, due to the prohibitive cost of collecting real-world extreme weather data, physics-based simulation has become a growing trend. Han et al. [4] proposed a weather simulation algorithm based on physical illumination models, synthesizing realistic rain and fog data on the KITTI dataset to effectively expand the diversity of training samples. A comparison of representative adverse weather datasets is presented in TABLE I.

TABLE I. TABLE I COMPARISON OF REPRESENTATIVE ADVERSE WEATHER DATASETS

Dataset	Weather Conditions	Year	Key Features
nuScenes	Mostly Clear/Cloudy, Light Rain	2020	Industry standard benchmark; lacks samples in extreme adverse weather.
SeeingThroughFog ^[6]	Different Density Fog, Snow, Rain	2020	Introduces gated camera; provides detailed visibility distance annotations.
CADC ^[9]	Heavy Snow, Light Snow, Slushy Roads	2021	Focuses on winter driving; includes cases of sensor occlusion by snow.
RADIATE ^[8]	Rain, Fog, Snow, Night	2021	First high-res mechanical scanning radar dataset; all-weather coverage.
Adverse-Weather ^[4]	Fog, Low-light, Sandstorm	2025	Physics-based synthetic data; used to address long-tail distribution.

III. ROBUST MULTI-MODAL FUSION PERCEPTION METHODOLOGIES

Faced with asymmetric sensor failures and severe data noise caused by adverse weather, simple sensor data concatenation is no longer sufficient to meet safety requirements. Recent research has shifted focus from traditional late fusion to deep fusion based on Unified Feature Spaces, Adaptive Weighting, and the Integration of Novel Enhanced Modalities.

A. Unified Representation Architectures and BEV Space Fusion

Traditional fusion methods typically rely on strict geometric projections, such as projecting LiDAR point clouds onto the image plane. This approach makes the system highly sensitive to spatiotemporal synchronization errors and prone to the collapse of the entire perception pipeline if a single modality fails. To address this issue, constructing a unified Bird's-Eye-View (BEV) representation space has become a mainstream trend.

The BEVFusion architecture proposed by Liu et al. [10] decouples camera and LiDAR features by establishing a shared BEV space. Unlike early methods such as PointPainting that rely on LiDAR point clouds as indices, BEVFusion allows each modality to extract features independently before fusion. Even in scenarios where LiDAR data is completely lost due to adverse weather, the system can maintain basic perception capabilities relying solely on camera features, significantly enhancing

system robustness. This architecture solves the spatial alignment problem under adverse weather by transforming sensor data from different views into a shared feature space. The schematic

of the BEV-based unified multi-modal fusion architecture is shown in Fig. 1.

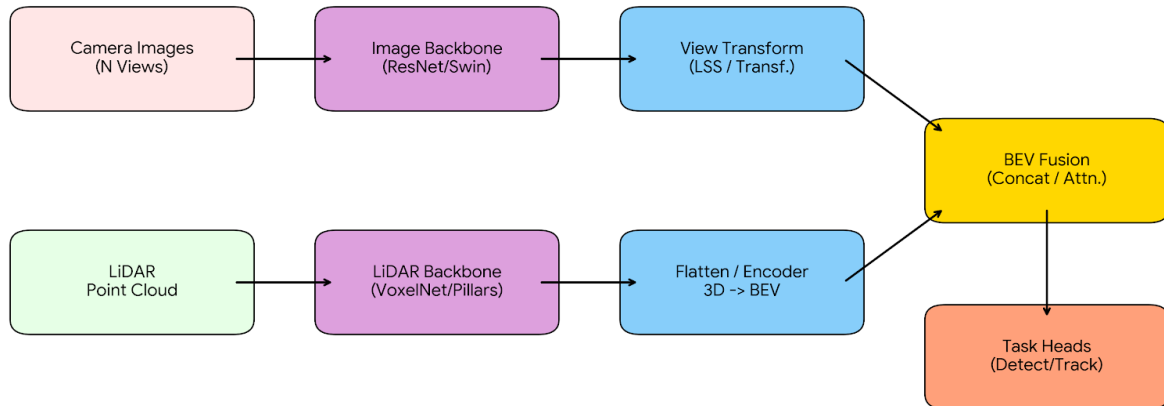


Fig. 1. Unified BEV Fusion Architecture

To further mitigate spatiotemporal sensor misalignment under adverse weather, Bai et al. ^[11] introduced **TransFusion**, which leverages the soft-attention mechanism of **Transformers**. This method uses LiDAR features as queries to find correspondences in image feature maps. This "soft association" mechanism can tolerate a certain degree of calibration error and automatically reduce the attention weights of blurred image regions. Similarly, Yang et al. ^[12] proposed **DeepInteraction**, which maintains modality-independent representations while performing multi-layer interactions, thereby maximizing the preservation of unique information from each sensor in extreme environments.

B. Adaptive Fusion and Domain Robustness Strategies

Since different sensors exhibit asymmetric failure under adverse weather—for instance, LiDAR fails in fog while radar remains effective—fusion strategies with fixed weights often yield suboptimal results. Therefore, introducing environment-

aware dynamic weight allocation mechanisms is a key measure to improve perception system robustness.

Sumalatha et al. ^[13] proposed a context-aware Adaptive Fusion Algorithm that dynamically adjusts the trust weights of each sensor by real-time monitoring of environmental conditions, such as light intensity and rainfall. Building on this, Palladin et al. ^[14] introduced a SOTA method, SAMFusion, which incorporates a distance-aware fusion mechanism. This approach prioritizes LiDAR features in close-range, high-confidence areas, while automatically increasing the weights of radar and gated cameras in long-range or high-scattering regions. This Transformer-based adaptive query mechanism effectively suppresses the interference of "ghost noise" on detection results. The network dynamically generates fusion weights for each modality based on input signal-to-noise ratios or entropy values to suppress sensor signals severely affected by weather. The schematic of the adaptive multi-modal fusion network is shown in Fig. 2.

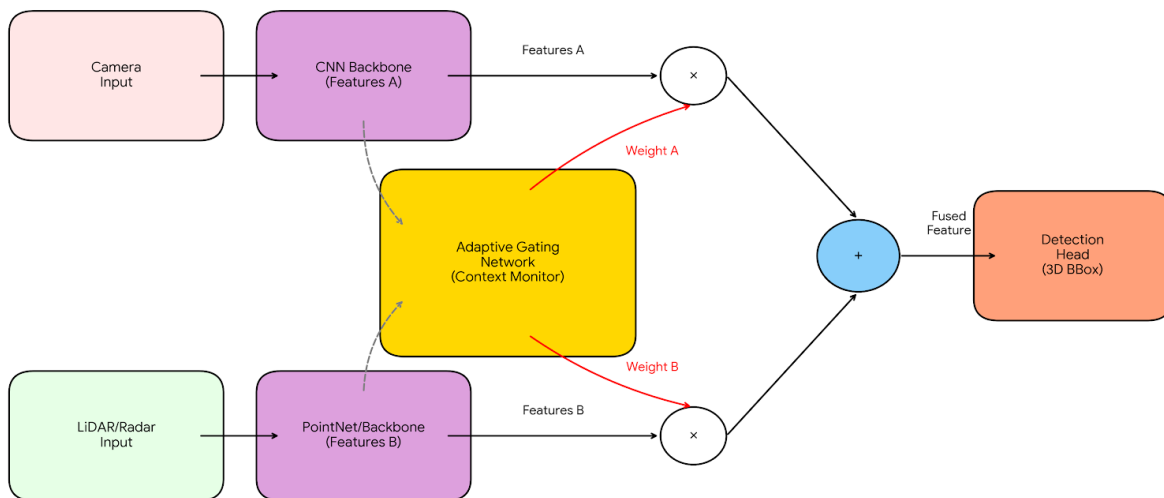


Fig. 2. Sensor-Adaptive Fusion Network Architecture

Beyond fusion strategies, addressing the Domain Shift caused by weather changes is also critical. Su et al. [15] pointed out that models suffer from "catastrophic forgetting" when transferring from clear weather to rain or snow. Their proposed D-FusionNet combines Domain-Incremental Learning, utilizing Gram matrices to measure feature distribution differences, enabling the model to continuously learn new weather scenarios without forgetting old knowledge. Additionally, Gupta et al. [16] proposed a Test-Time Adaptation (TTA) strategy, which employs adaptive augmentation during the inference phase to handle sudden adverse weather without retraining the model.

C. Integration of Enhanced Modalities

To breakthrough the physical limits of conventional sensors, integrating novel modalities with penetration capabilities is becoming a new research hotspot.

- **Thermal Fusion:** Addressing unlit nighttime and dense smoke scenarios, Nawaz et al. [17] verified the irreplaceability of Thermal Cameras. By fusing thermal data with RGB and LiDAR, the system can effectively identify pedestrians and animals based on thermal radiation signatures, filling the blind spots of visual perception in darkness and smoke.
- **Gated Cameras and De-scattering:** To tackle backscatter in dense fog, Bijelic [6] and Gruber et al. [8] utilized Gated Cameras with time-slicing technology to filter out near-field reflection noise and directly acquire depth and intensity information of target objects, significantly extending detection range in fog.
- **Radar-Enhanced Perception:** Considering the balance between cost and robustness, Nabati et al. [18] proposed CenterFusion, which leverages the radial velocity and depth information of millimeter-wave radar to assist monocular cameras. This low-cost "Radar+Vision" solution demonstrates remarkable stability in rainy conditions, proving that radar can serve as an effective alternative to LiDAR in specific scenarios.

D. Downstream Tasks: Robust Multi-Object Tracking

Perception uncertainty directly propagates to downstream tracking tasks. Under adverse weather, jitter in detection bounding boxes leads to frequent identity switches (ID Switch) in Multi-Object Tracking (MOT). Han et al. [4] proposed a Progressive Association Strategy tailored for adverse weather, which prioritizes spatial consistency of point clouds for matching before confirming with visual appearance features. Combining this with physics-based weather data augmentation, the method significantly reduces tracking errors under rain and fog interference, ensuring trajectory continuity.

In summary, while BEV- and Transformer-based sensor fusion architectures demonstrate superior performance in addressing spatiotemporal alignment and feature enhancement under adverse weather, these gains often come at the cost of substantial computational overhead and model complexity. Current academic research predominantly focuses on accuracy breakthroughs on benchmarks, often overlooking the resource constraints of in-vehicle embedded platforms and rigorous safety standards. Bridging the engineering gap from algorithm

prototypes to production-level systems remains a critical challenge in automotive development.

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IV. ENGINEERING CHALLENGES AND REAL-WORLD DEPLOYMENT

While multi-sensor fusion algorithms have achieved significant performance gains on adverse weather datasets, translating these achievements into production-level autonomous driving systems faces severe engineering challenges. Key factors constraining the commercial deployment of adverse weather perception technologies include computational efficiency constraints for complex denoising algorithms, functional safety certification in extreme environments, and weather-induced system integration failures.

A. Computational Efficiency and Real-Time Constraints

To mitigate interference from rain and fog, modern perception architectures often integrate complex denoising modules or Transformer-based spatiotemporal alignment mechanisms. These additions introduce substantial computational overhead and pose severe tests for in-vehicle edge computing platforms, thereby constraining the computational efficiency and real-time performance of autonomous driving systems.

Compared to clear weather scenarios, robust perception models for adverse weather typically require processing additional feature dimensions or executing high-latency attention computations. Unlike high-performance servers used in laboratory settings, automotive-grade chips are subject to strict power consumption and thermal envelopes. Consequently, complex fusion models often result in inference latencies exceeding 100 ms, making it difficult to meet the strict requirements for 10 Hz–30 Hz perception update rates and the decision-making loop.

To address these issues, model lightweighting is essential. To deploy high-precision models on edge devices, model compression techniques must be adopted. Research indicates that by utilizing integer quantization techniques, it is possible to significantly reduce memory bandwidth usage and inference latency while maintaining model accuracy comparable to floating-point baselines [19]. This efficiency is a prerequisite for deploying complex, high-dimensional weather-robust models on resource-constrained edge devices.

Furthermore, energy efficiency balance is critical. Electric vehicles inherently face range attenuation challenges in low-temperature or heavy rain conditions; high-compute perception systems further deplete battery energy, impacting driving range.

Therefore, the academic community has begun exploring Weather-Aware Adaptive Perception strategies. These systems activate high-energy denoising or fusion modules only when adverse weather is detected, automatically switching to a lightweight mode in favorable weather, thereby achieving an optimal balance of energy efficiency across all-weather scenarios [20].

B. Functional Safety and SOTIF Certification

In adverse weather, perception system failures often stem from physical limitations rather than hardware faults, rendering traditional safety standards insufficient. This necessitates safety standards and certification requirements that are better suited for autonomous driving systems.

Addressing Weather Limitations with SOTIF(Safety of the Intended Functionality): Traditional standards like ISO 26262 primarily address hardware faults in electrical and electronic systems (e.g., sensor circuit breaks) but fail to cover perception performance degradation caused by environmental factors such as blinding snow or glare. To address the risk of failure in unknown scenarios, SOTIF (ISO 21448) has become a mandatory standard for autonomous driving systems[21]. SOTIF requires that the system maintains false positive risks within an acceptable range even under specific "triggering conditions."

Quantitative Decomposition of Adverse Weather Metrics: To pass certification, qualitative "robustness" must be translated into quantitative engineering metrics. Recent studies propose deriving minimum performance requirements for perception systems based on the road adhesion coefficient under different weather conditions. For example, the goal of "safe driving in fog" can be decomposed into specific Key Performance Indicators (KPIs), such as a defined minimum detection range and maximum allowable false-negative rate when visibility is below 50 m [22]. This quantitative verification framework is a necessary path for algorithms to evolve from prototypes to products and is an effective measure for the continuous evolution of autonomous driving in the automotive industry.

C. System Integration and Fault Tolerance

Real-world adverse weather not only affects signal transmission but also physically erodes sensor hardware, leading to failures at the system integration level, which requires specific coping strategies.

Environment-Induced Calibration Drift: Extreme temperature differences can cause thermal expansion and contraction of sensor mounts, leading to extrinsic calibration drift; meanwhile, mud splashes or ice coverage can cause severe signal attenuation. A robust fusion system must include Online Self-Calibration and Blockage Detection modules specifically designed for adverse weather [23].

Fault Isolation and Degradation Strategies: When force majeure events (such as torrential rain) cause the complete failure of a specific modality, the system must possess Fault Isolation capabilities to prevent noise data from contaminating the entire fusion center. In engineering practice, Degradation Strategies are commonly employed. For instance, in the event of visual failure, the system automatically switches to a radar-dominant "Limp-home Mode," relying on the remaining sensors to maintain minimum safe driving functions [23].

V. CONCLUSION AND FUTURE TRENDS

A. Conclusion

This paper has presented a systematic review of multi-sensor fusion perception technologies tailored for adverse weather conditions in autonomous driving. Research indicates that extreme meteorological phenomena—such as rain, snow, and fog—induce asymmetric physical failures in individual sensors, notably LiDAR blinding due to scattering and visual degradation in cameras [5][6]. To address this challenge, perception architectures are evolving from traditional late fusion toward deep fusion based on Unified BEV Feature Spaces [10], incorporating the soft-attention mechanisms of Transformers to resolve spatiotemporal misalignment [11]. Adaptive Weight Assignment [14] and Domain-Incremental Learning [15] strategies have emerged as effective means to enhance the robustness of perception models under adverse weather conditions. However, the commercial deployment of these algorithms, particularly in adverse weather scenarios, is constrained by the conflict between onboard computational resources and real-time requirements, the lack of Safety of the Intended Functionality (SOTIF) certification, and system integration failures induced by environmental factors.

B. Future Outlook

Despite significant progress in existing methods, achieving all-weather L5 autonomous driving still faces severe challenges. Future research trends will focus on the following two directions.

1) *Next-Generation Sensor Hardware: 4D Imaging Radar.* Current millimeter-wave radars cannot yet match LiDAR in angular resolution. However, with the maturation of 4D Imaging Radar technology, it can generate high-density point clouds containing elevation information while retaining all-weather penetration characteristics [24]. Introducing 4D radar as a primary sensor into fusion frameworks holds the promise of significantly solving high-precision perception challenges in adverse weather.

2) *Beyond-Line-of-Sight Perception: V2X Cooperation.* Under extreme adverse weather conditions such as torrential rain or sandstorms, the physical limits of single-vehicle perception are difficult to overcome. V2X Cooperative Perception technology enables vehicles to utilize sensor data from Roadside Units (RSUs) or other connected vehicles, realizing "see-through" occlusion capabilities and beyond-line-of-sight perception [25]. Constructing a cloud-edge-vehicle collaborative fusion network based on V2X will be an effective solution for coping with extreme catastrophic weather in the future.

Furthermore, research on perception in adverse weather should not be limited to pursuing higher detection accuracy; instead, it must evolve toward a paradigm that balances "high performance with high reliability."

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